

# Qatar-2b

R-1105

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_Qatar-2\_160412 · created 2026-07-30T20:07:19+00:00

## BENCHMARK: NON-DETECTION

### Results

|                                                   |                                   |
|---------------------------------------------------|-----------------------------------|
| Mid-Transit Time (Tmid)                           | 2457490.88199 +/- 0.00026 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.002 +/- 0.065                   |
| Transit depth (Rp/Rs)^2                           | 0.00026 +/- 0.021 [%]             |
| Orbital Inclination (inc)                         | 87.91 +/- 1.71                    |
| Airmass coefficient 1 (a1)                        | 1.71 +/- 1.02                     |
| Airmass coefficient 2 (a2)                        | -0.34 +/- 0.2                     |
| Scatter in the residuals of the lightcurve fit is | 59.94 %                           |
| Best Comparison Star                              | None                              |
| Optimal Aperture                                  | 3.0                               |
| Optimal Annulus                                   | 7.5                               |
| Transit Duration (day)                            | 0.0665 +/- 0.0043                 |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                            |
|---------------------------|--------------------------------------------|
| Rp/R■ fit vs published    | 0.002 ± 0.065 vs 0.16327 (deviation 2.48σ) |
| Duration fit vs published | 0.0665 d vs 0.0767 d (deviation 2.36σ)     |
| Mid-time O–C              | 0.3 min                                    |
| Depth SNR                 | 0.0                                        |
| Residual scatter          | 59.94 %                                    |

### Light curve

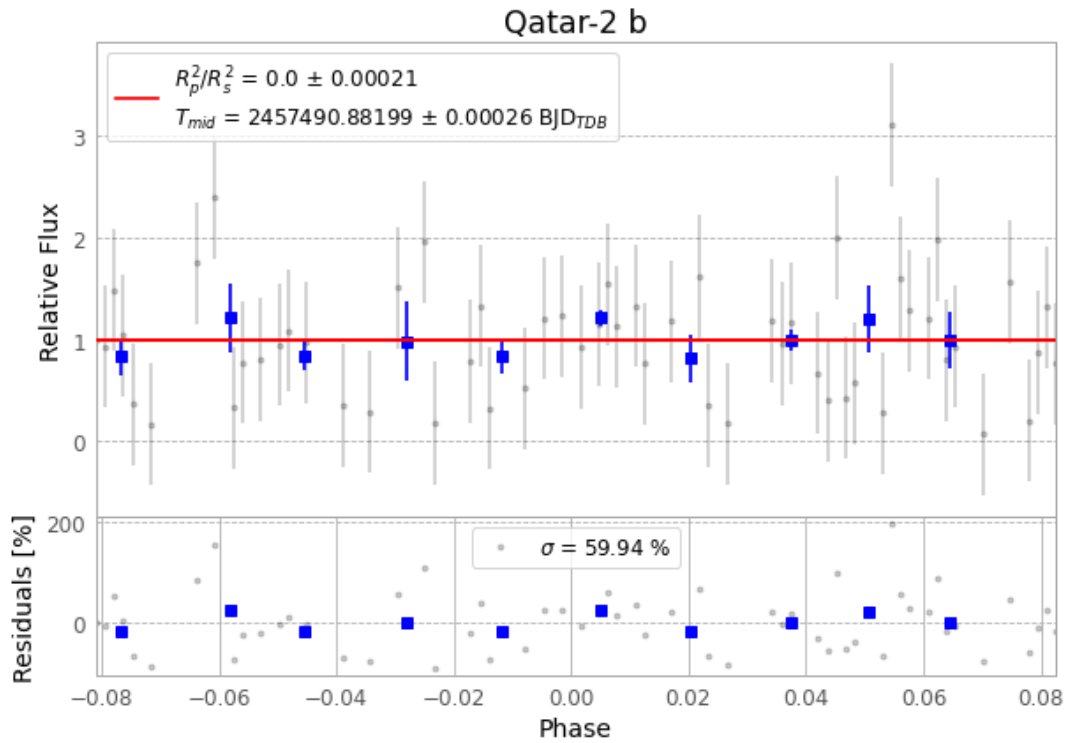


Figure 1 — FinalLightCurve\_Qatar-2 b\_2016-04-11.png (SHA-256 8765705189317f9f...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [142, 433], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 de96879339892aa1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit ba3ef8b

## Data provenance

111 FITS frames (73 MB), Qatar-2160412062112.FITS → Qatar-2160412115112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-160412.log (1e0c6180490b...), FinalLightCurve\_Qatar-2 b\_2016-04-11.png (876570518931...), FinalLightCurve\_Qatar-2 b\_2016-04-11.csv (5bdeb63eb5d2...)

## Compute resources

Wall time 250.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 20:07Z.